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Airport (20_582) Homework 3

Problem 1

The characteristics of a new aircraft are given in Table B-1. In domestic route service, the applicable regulations require only a 0.75-h fuel reserve for this aircraft. The aircraft has an average route speed of 500 mi/h and an average fuel burn of 13 lb/mi.

Characteristic	Value (lb)
Maximum ramp weight	141000
Maximum structural takeoff weight	140000
Maximum structural landing weight	128000
Zero-fuel weight	112000
Operating empty weight	73274
Fuel capacity	39162
Maximum structural payload	38726
Maximum passenger capacity	27000
Maximum cargo hold capacity	12500

Table 1: Preliminary aircraft characteristics

- (a) Plot payload versus range for this aircraft.
- (b) Determine the takeoff weight of this aircraft for a 1200-mi trip at maximum payload.

Solution

The reserve flight time is:

$$t_{reserve} = 0.75 \text{ h}$$

Since the aircraft speed is:

$$V = 500 \text{ mi/h}$$

the reserve distance becomes:

$$R_{reserve} = 0.75 \times 500 = 375 \text{ mi}$$

Given the fuel burn rate:

$$13 \text{ lb/mi}$$

the reserve fuel weight is:

$$W_{fuel, reserve} = 375 \times 13 = 4875 \text{ lb}$$

The takeoff weight relation is:

$$W_{TO} = W_{OEI} + W_{payload} + W_{fuel}$$

where:

$$W_{OEI} = 73274 \text{ lb}$$

and the maximum payload is:

$$W_{payload, max} = 38726 \text{ lb}$$

(a) Payload–Range Diagram

At maximum payload:

$$W_{fuel} = 140000 - 73274 - 38726$$

$$W_{fuel} = 28000 \text{ lb}$$

The usable trip fuel is:

$$W_{fuel, trip} = 28000 - 4875 = 23125 \text{ lb}$$

Thus, the range is:

$$R = \frac{23125}{13} \approx 1779 \text{ mi}$$

For maximum fuel capacity:

$$W_{fuel, max} = 39162 \text{ lb}$$

The payload allowed by MTOW becomes:

$$W_{payload} = 140000 - 73274 - 39162$$

$$W_{payload} = 27564 \text{ lb}$$

The corresponding trip fuel is:

$$39162 - 4875 = 34287 \text{ lb}$$

Therefore:

$$R = \frac{34287}{13} \approx 2637 \text{ mi}$$

Range (mi)	Payload (lb)
0	38726
1779	38726
2637	27564
2637	0

Table 2: Payload–range characteristic points

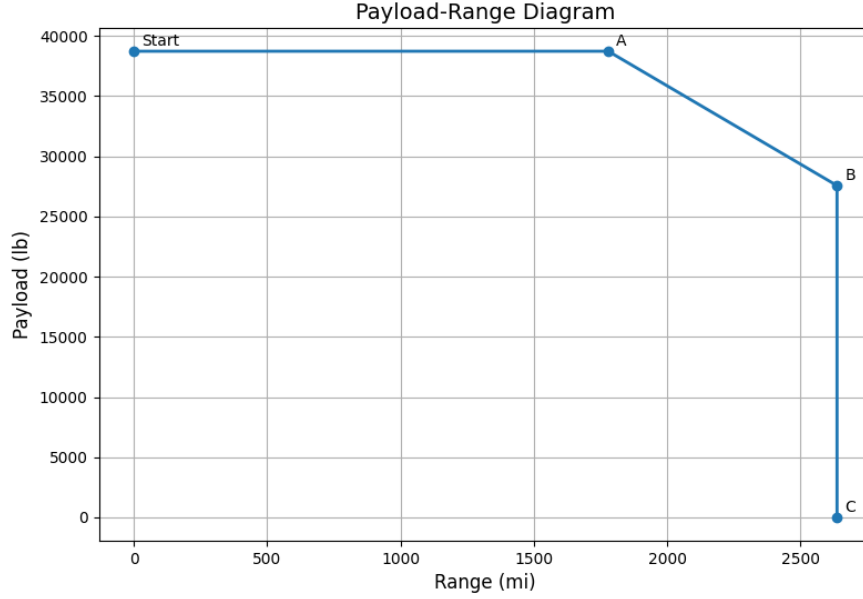


Figure 1: Payload–range diagram

(b) Takeoff Weight for a 1200-mi Trip

For a trip distance of:

$$R = 1200 \text{ mi}$$

the required trip fuel is:

$$W_{fuel,trip} = 1200 \times 13 = 15600 \text{ lb}$$

Adding reserve fuel:

$$W_{fuel,total} = 15600 + 4875 = 20475 \text{ lb}$$

The takeoff weight becomes:

$$W_{TO} = 73274 + 38726 + 20475$$

$$W_{TO} = 132475 \text{ lb}$$

Therefore, the required takeoff weight is:

$$\boxed{132475 \text{ lb}}$$

Problem 2

An aircraft has the performance characteristics given in Table B-2 in normal route operation. In normal use this aircraft will consume 11000 lb of fuel in 1 h of flight. The normal cruising speed is 550 mi/h. Generally, alternate destinations are located within 100 mi of planned destination, and regulations require sufficient fuel to fly to the planned destination, fly to an alternate destination, and fly for 1.25 h at the alternate destination.

Aircraft Data

Characteristic	Value (lb)
Maximum ramp weight	191000
Maximum structural takeoff weight	190000
Maximum structural landing weight	160000
Zero-fuel weight	140000
Operating empty weight	100000
Fuel capacity	55000
Maximum structural payload	45000
Maximum passenger capacity	39000
Maximum cargo hold capacity	15000

Table 3: Aircraft performance characteristics

The fuel consumption is:

$$11000 \text{ lb/h}$$

The cruise speed is:

$$550 \text{ mi/h}$$

Thus, the fuel burn per mile is:

$$\frac{11000}{550} = 20 \text{ lb/mi}$$

The reserve fuel consists of:

- Fuel to fly 100 mi to alternate airport:

$$100 \times 20 = 2000 \text{ lb}$$

- Fuel for 1.25 h holding:

$$1.25 \times 11000 = 13750 \text{ lb}$$

Therefore, total reserve fuel is:

$$2000 + 13750 = 15750 \text{ lb}$$

(a) Maximum Passenger Payload with 15000 lb Cargo

The maximum payload allowed by the zero-fuel-weight limit is:

$$140000 - 100000 = 40000 \text{ lb}$$

Since 15000 lb of cargo is carried:

$$\text{Passenger payload} = 40000 - 15000 = 25000 \text{ lb}$$

Assuming 200 lb per passenger:

$$\frac{25000}{200} = 125$$

Therefore:

25000 lb passenger payload

or

125 passengers

(b) Fuel Required for a 2000-mi Trip

Trip fuel:

$$2000 \times 20 = 40000 \text{ lb}$$

Required takeoff fuel:

$$40000 + 15750 = 55750 \text{ lb}$$

Including 1000 lb taxi fuel:

$$55750 + 1000 = 56750 \text{ lb}$$

Therefore:

56750 lb

Since the fuel capacity is only 55000 lb, this flight is not feasible nonstop under the given assumptions.

(c) Additional Airborne Time Before Landing

The aircraft carries 180 passengers and no cargo except baggage.

Assuming 200 lb per passenger:

$$180 \times 200 = 36000 \text{ lb}$$

Zero-fuel weight becomes:

$$100000 + 36000 = 136000 \text{ lb}$$

Maximum allowable fuel at takeoff:

$$190000 - 136000 = 54000 \text{ lb}$$

Fuel burned in the first 30 minutes:

$$0.5 \times 11000 = 5500 \text{ lb}$$

Remaining fuel:

$$54000 - 5500 = 48500 \text{ lb}$$

Aircraft weight after 30 minutes:

$$136000 + 48500 = 184500 \text{ lb}$$

The maximum landing weight is:

$$160000 \text{ lb}$$

Excess weight:

$$184500 - 160000 = 24500 \text{ lb}$$

This fuel must be burned before landing.

Additional airborne time required:

$$\frac{24500}{11000} = 2.23 \text{ h}$$

Therefore:

$$\boxed{2.23 \text{ h}}$$

which is approximately:

$$\boxed{2 \text{ h } 14 \text{ min}}$$

(d) Payload–Range Diagram

The payload–range diagram is constructed using the aircraft weight limitations and fuel capacity constraints.

Point A: Maximum Payload Condition

The maximum payload is limited by the zero-fuel weight:

$$W_{\text{payload,max}} = 140000 - 100000 = 40000 \text{ lb}$$

Thus point A is:

$$A = (0, 40000)$$

Point B: Maximum Payload at MTOW

At maximum payload, the takeoff weight is:

$$W_{TO} = 100000 + 40000 + W_f$$

Using the maximum takeoff weight:

$$190000 = 100000 + 40000 + W_f$$

Thus:

$$W_f = 50000 \text{ lb}$$

The reserve fuel requirement is:

$$W_{res} = 15750 \text{ lb}$$

Therefore usable cruise fuel is:

$$W_{cruise} = 50000 - 15750 = 34250 \text{ lb}$$

Since fuel burn is:

$$20 \text{ lb/mi}$$

the corresponding range is:

$$R_B = \frac{34250}{20} = 1712.5 \text{ mi}$$

Thus point B is:

$$B = (1712.5, 40000)$$

Point C: Maximum Fuel Capacity

At maximum fuel capacity:

$$W_f = 55000 \text{ lb}$$

The allowable payload becomes:

$$W_{payload} = 190000 - 100000 - 55000 = 35000 \text{ lb}$$

Usable cruise fuel is:

$$55000 - 15750 = 39250 \text{ lb}$$

Thus the range is:

$$R_C = \frac{39250}{20} = 1962.5 \text{ mi}$$

Therefore point C is:

$$C = (1962.5, 35000)$$

Point D: Ferry Range

Below point C, payload can continue decreasing while fuel remains fixed at maximum fuel capacity. Therefore the range remains constant in this simplified model.

Thus the ferry-range point is:

$$D = (1962.5, 0)$$

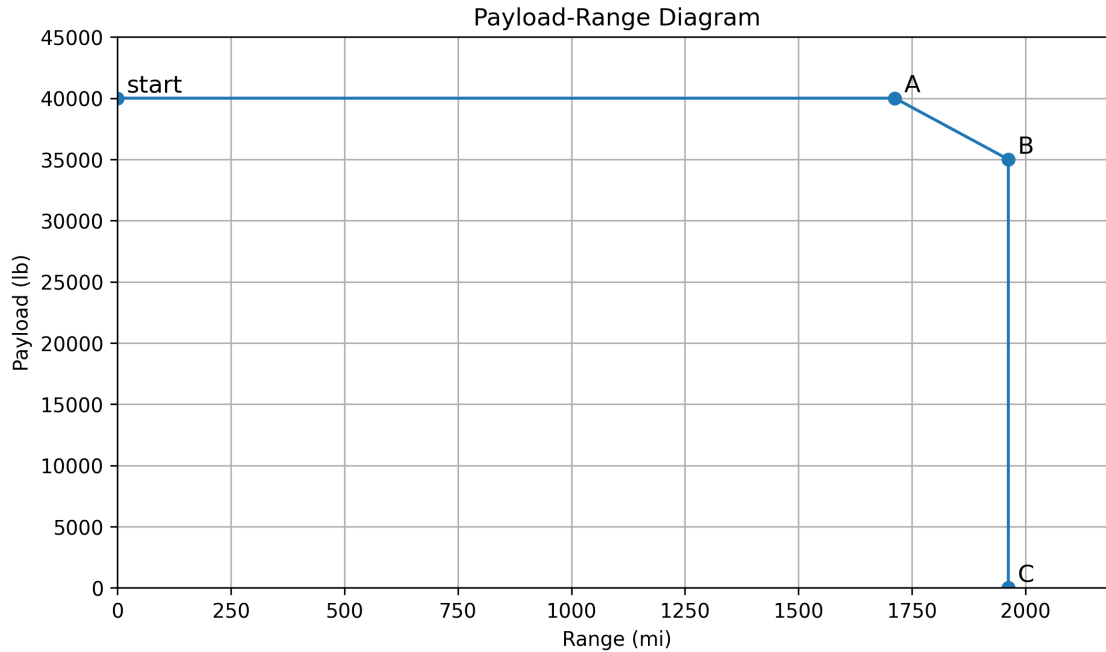


Figure 2: Payload versus route range diagram

Problem 3

A turbine-powered aircraft has the following runway performance characteristics:

- For a normal takeoff, the liftoff distance is 6500 ft, and the aircraft reaches a height of 35 ft above the end of the runway at a distance of 7000 ft from the runway threshold.
- For an engine-failure takeoff, the liftoff distance is 7200 ft, and the aircraft reaches a height of 35 ft above the end of the runway at a distance of 7900 ft from the runway threshold.
- For an aborted takeoff, the accelerate-stop distance is 9700 ft.
- For landing, the stop distance is 4600 ft.

Determine the minimum FAR runway length requirements for this aircraft, indicating:

- Length of the full-strength pavement
- Length of the stopway
- Length of the clearway

Assume the runway is to be used by this aircraft in both directions.

Solution

1. Normal Takeoff

Given:

$$LOD_1 = 6500 \text{ ft}$$

$$D_{35} = 7000 \text{ ft}$$

The takeoff distance is:

$$TOD_1 = 1.15 \times D_{35}$$

$$TOD_1 = 1.15 \times 7000 = 8050 \text{ ft}$$

The maximum clearway length is:

$$CL_{1,\max} = \frac{1}{2} (TOD_1 - 1.15 \times LOD_1)$$

$$CL_{1,\max} = \frac{1}{2} (8050 - 1.15 \times 6500)$$

$$CL_{1,\max} = \frac{1}{2} (8050 - 7475)$$

$$CL_{1,\max} = 287.5 \text{ ft}$$

Therefore, the takeoff run required is:

$$TOR_1 = TOD_1 - CL_{1,\max}$$

$$TOR_1 = 8050 - 287.5$$

$$TOR_1 = 7762.5 \text{ ft}$$

2. Engine-Failure Takeoff

Given:

$$LOD_2 = 7200 \text{ ft}$$

$$D_{35} = 7900 \text{ ft}$$

For engine-failure takeoff:

$$TOD_2 = D_{35} = 7900 \text{ ft}$$

The maximum clearway length is:

$$CL_{2,\max} = \frac{1}{2} (TOD_2 - LOD_2)$$

$$CL_{2,\max} = \frac{1}{2}(7900 - 7200)$$

$$CL_{2,\max} = 350 \text{ ft}$$

Thus, the takeoff run required becomes:

$$TOR_2 = TOD_2 - CL_{2,\max}$$

$$TOR_2 = 7900 - 350$$

$$TOR_2 = 7550 \text{ ft}$$

3. Aborted Takeoff

Given:

$$DAS = 9700 \text{ ft}$$

4. Landing Requirement

Given stopping distance:

$$SD = 4600 \text{ ft}$$

Using FAR landing requirement:

$$LD = \frac{SD}{0.6}$$

$$LD = \frac{4600}{0.6}$$

$$LD = 7667 \text{ ft}$$

5. Runway Length Determination

The required full runway length is:

$$FL = \max(TOD_1, TOD_2, DAS, LD)$$

$$FL = \max(8050, 7900, 9700, 7667)$$

$$FL = 9700 \text{ ft}$$

The required full-strength pavement length is:

$$FS = \max(TOR_1, TOR_2, LD)$$

$$FS = \max(7762.5, 7550, 7667)$$

$$FS = 7762.5 \text{ ft}$$

The required stopway length is:

$$SW = DAS - \max(TOR_1, TOR_2, LD)$$

$$SW = 9700 - 7762.5$$

$$SW = 1937.5 \text{ ft}$$

The required clearway length is:

$$CL = \min(FL - DAS, CL_{1,\max}, CL_{2,\max})$$

$$CL = \min(9700 - 9700, 287.5, 350)$$

$$CL = 0 \text{ ft}$$

Final Answers

Full runway length = 9700 ft

Full-strength pavement length = 7762.5 ft

Stopway length = 1937.5 ft

Clearway length = 0 ft

Problem 4

An airline is considering the use of a particular aircraft on a non-stop route of 1230 mi between two airports. The nearest alternate airport at either destination is 260 mi away. The time fuel reserves are included in the item listed as “typical operating empty weight plus time fuel reserves” in Table B-3. Table B-4 indicates the takeoff requirements for this aircraft.

The airport under study has a runway of 8000 ft available with a maximum difference in centerline elevation of 15 ft. The airport elevation is 500 ft above mean sea level (AMSL), and the mean daily highest temperature in the hottest month is 90°F.

- (a) Find the maximum allowable takeoff weight of this aircraft at the study airport.
- (b) If the aircraft takes off from the other airport with a takeoff weight of 338,000 lb with the maximum allowable payload and minimum fuel, find:
 - (a) the maximum allowable payload at the origin airport,
 - (b) the landing weight,
 - (c) and the landing runway requirements at the destination airport.

Table 4: Landing Runway Requirements

Maximum Allowable Landing Weight, 1000 lb							
Airport elevation, ft							
Temp. (°F)	0	1000	2000	3000	4000	5000	6000
80	363.5	363.5	363.5	363.5	363.5	363.5	363.5
85	363.5	363.5	363.5	363.5	363.5	363.5	359.8
90	363.5	363.5	363.5	363.5	363.5	363.5	350.6
95	363.5	363.5	363.5	363.5	363.5	354.6	341.7
100	363.5	363.5	363.5	363.5	358.4	345.5	332.8

Landing Runway Length, 1000 ft							
Airport elevation, ft							
Weight (1000 lb)	0	1000	2000	3000	4000	5000	6000
280	5.44	5.57	5.71	5.85	6.00	6.15	6.31
290	5.59	5.73	5.87	6.02	6.17	6.33	6.49
300	5.75	5.89	6.04	6.19	6.35	6.51	6.68
310	5.90	6.05	6.20	6.36	6.52	6.69	6.86
320	6.06	6.21	6.37	6.53	6.70	6.87	7.05
330	6.21	6.37	6.54	6.70	6.87	7.05	7.23
340	6.37	6.53	6.69	6.86	7.03	7.21	7.40
350	6.51	6.66	6.83	7.00	7.19	7.37	7.57
360	6.64	6.81	6.98	7.16	7.35	7.54	7.75
370	6.77	6.96	7.14	7.33	7.52	7.71	7.92

Maximum structural takeoff weight = 408,900 lb

Maximum structural landing weight = 363,500 lb

Typical operating weight plus time fuel reserves = 263,300 lb

Average fuel consumption = 30 lb/mi

Typical maximum passenger load = 54,000 lb

Maximum structural payload = 99,660 lb

Fuel capacity = 21,762 gal

0.1 Solution

Given data:

$$R = 1230 \text{ mi}$$

$$R_{alt} = 260 \text{ mi}$$

$$h = 500 \text{ ft}$$

$$T = 90^\circ F$$

$$L_{avail} = 8000 \text{ ft}$$

The runway centerline elevation difference is:

$$\Delta h = 15 \text{ ft}$$

Therefore, the runway slope is:

$$S = \frac{15}{8000} \times 100 = 0.1875\%$$

Since the slope is very small, no correction is required.

Table 5: Takeoff Runway Requirements

		Airport elevation, ft						
Temp. (°F)		0	1000	2000	3000	4000	5000	6000
Maximum Allowable Takeoff Weight, 1000 lb								
80		408.9	400.6	392.9	382.8	369.1	356.2	343.9
85		409.0	400.6	386.0	372.3	359.4	347.1	335.4
90		403.6	388.8	374.9	362.0	349.7	338.1	326.9
95		391.1	377.3	364.2	352.0	340.3	329.2	318.4
100		379.6	366.4	354.1	342.4	331.2	320.4	309.9

		Reference Factor						
Temp. (°F)		0	1000	2000	3000	4000	5000	6000
80		46.0	48.2	50.5	53.4	57.0	61.0	65.4
85		46.3	48.5	51.8	55.2	59.0	63.1	67.8
90		47.2	50.3	53.6	57.2	61.1	65.5	70.4
95		48.8	52.0	55.4	59.2	63.3	68.0	73.2
100		50.5	53.8	57.4	61.4	65.8	70.7	76.3

		Takeoff Runway Length, 1000 ft				
		Reference factor				
Weight (1000 lb)		40	50	60	70	80
320		4.23	5.28	6.39	7.50	8.56
330		4.44	5.58	6.78	7.98	9.09
340		4.67	5.89	7.18	8.46	9.63
350		4.90	6.21	7.60	8.95	10.16
360		5.13	6.55	8.03	9.45	10.68
370		5.38	6.89	8.46	9.94	—
380		5.63	7.25	8.91	10.44	—
390		5.89	7.63	9.36	10.92	—
400		6.17	8.01	9.82	—	—

(a) Maximum Allowable Takeoff Weight

Using Table B-4 for a temperature of $90^{\circ}F$:

Elevation (ft)	Maximum Takeoff Weight (1000 lb)
0	403.6
1000	388.8

Interpolating for 500 ft elevation:

$$W_{TO,max} = 403.6 + \frac{500}{1000}(388.8 - 403.6)$$

$$W_{TO,max} = 396.2 \text{ klb}$$

Thus,

$$W_{TO,max} \approx 396,200 \text{ lb}$$

(b) Payload, Landing Weight, and Landing Runway Requirement

The total distance including the alternate airport is:

$$R_{tot} = 1230 + 260 = 1490 \text{ mi}$$

The average fuel consumption is:

$$30 \text{ lb/mi}$$

Therefore, the required fuel weight is:

$$W_f = 1490 \times 30$$

$$W_f = 44,700 \text{ lb}$$

The operating weight plus reserves is:

$$W_{OE} = 263,300 \text{ lb}$$

The payload corresponding to the maximum allowable takeoff weight is:

$$W_P = W_{TO,max} - W_{OE} - W_f$$

$$W_P = 396,200 - 263,300 - 44,700$$

$$W_P = 88,200 \text{ lb}$$

However, the aircraft limitations are:

$$\text{Typical maximum passenger load} = 54,000 \text{ lb}$$

$$\text{Maximum structural payload} = 99,660 \text{ lb}$$

Therefore, the allowable payload is limited to:

$$W_P = 54,000 \text{ lb}$$

Landing Weight

The actual takeoff weight becomes:

$$W_{TO} = 263,300 + 54,000 + 44,700$$

$$W_{TO} = 362,000 \text{ lb}$$

Fuel burned during the trip from origin to destination:

$$W_{burn} = 1230 \times 30$$

$$W_{burn} = 36,900 \text{ lb}$$

Therefore, the landing weight is:

$$W_L = 362,000 - 36,900$$

$$W_L = 325,100 \text{ lb}$$

Thus,

$$\boxed{W_L = 325,100 \text{ lb}}$$

This value is below the maximum structural landing weight:

$$325,100 < 363,500 \text{ lb}$$

Therefore, the landing weight is acceptable.

Landing Runway Requirement

Using Table B-3:

For 320 klb:

$$L_{320} = 6.06 + \frac{500}{1000}(6.21 - 6.06)$$

$$L_{320} = 6.135 \text{ kft}$$

For 330 klb:

$$L_{330} = 6.21 + \frac{500}{1000}(6.37 - 6.21)$$

$$L_{330} = 6.29 \text{ kft}$$

Interpolating for 325.1 klb:

$$L = 6.135 + \frac{325.1 - 320}{330 - 320}(6.29 - 6.135)$$

$$L \approx 6.21 \text{ kft}$$

Therefore,

$$\boxed{L_{\text{landing}} \approx 6210 \text{ ft}}$$